Fail2Ban Lab (Ubuntu)

What is Fail2Ban?

Fail2Ban is a security tool that monitors log files and automatically blocks IP addresses that show malicious behavior such as repeated failed login attempts.

Working

```
Attacker


Multiple Failed Logins


Log File


Fail2Ban


iptables/UFW Rule


IP Blocked
```

Lab Objective

- Install Fail2Ban
- Protect SSH from brute-force attacks
- Monitor logs
- Automatically ban attacker IPs

Step 1: Install Fail2Ban

```
sudo apt update

sudo apt install fail2ban -y
```

Verify:

```
fail2ban-client --version
```

Check service:

```
sudo systemctl status fail2ban
```

Step 2: Enable and Start Service

```
sudo systemctl enable fail2ban

sudo systemctl start fail2ban
```

Verify:

```
sudo systemctl is-active fail2ban
```

Expected:

```
active
```

Step 3: Create Local Configuration

Never edit:

```
/etc/fail2ban/jail.conf
```

Create:

```
sudo cp /etc/fail2ban/jail.conf \
/etc/fail2ban/jail.local
```

Step 4: Configure SSH Protection

Edit:

```
sudo nano /etc/fail2ban/jail.local
```

Find:

```
[sshd]
```

Modify:

```
[sshd]
enabled = true
port = ssh
logpath = %(sshd_log)s
maxretry = 3
findtime = 300
bantime = 600
```

Meaning:

Parameter	Description
enabled	Enable SSH monitoring
maxretry=3	Ban after 3 failed attempts
findtime=300	Within 5 minutes

bantime=600	Ban for 10 minutes

Step 5: Restart Fail2Ban

```
sudo systemctl restart fail2ban
```

Verify:

```
sudo fail2ban-client status
```

Output:

```
Jail list: sshd
```

Step 6: Check SSH Jail

```
sudo fail2ban-client status sshd
```

Example:

```
Status for the jail: sshd
Currently failed: 0
Currently banned: 0
```

Step 7: Simulate Brute Force Attack

From Kali:

```
ssh fakeuser@SERVER_IP
```

Enter wrong password multiple times.

Or:

```
hydra -l root -P passwords.txt ssh://SERVER_IP
```

After exceeding retries:

```
IP gets banned
```

Step 8: Verify Banned IP

```
sudo fail2ban-client status sshd
```

Example:

```
Currently banned: 1

Banned IP list:
192.168.1.10
```

Step 9: Check Firewall Rule

If iptables is used:

```
sudo iptables -L -n
```

You will see:

```
f2b-sshd
```

Chain created by Fail2Ban.

Step 10: Unban IP

```
sudo fail2ban-client set sshd unbanip 192.168.1.10
```

Verify:

```
sudo fail2ban-client status sshd
```

Step 11: View Logs

```
sudo tail -f /var/log/fail2ban.log
```

Example:

```
Ban 192.168.1.10
```

Important Commands

Service Status

```
sudo systemctl status fail2ban
```

List Jails

```
sudo fail2ban-client status
```

SSH Jail Status

```
sudo fail2ban-client status sshd
```

Restart

```
sudo systemctl restart fail2ban
```

Unban IP

```
sudo fail2ban-client set sshd unbanip IP_ADDRESS
```

Architecture

```
SSH Server


Auth Log
(/var/log/auth.log)


Fail2Ban


iptables/UFW


Block Attacker IP
```

Example Attack Flow

```
Attacker IP:
192.168.1.10

Attempt 1    Wrong Password
Attempt 2    Wrong Password
```

```
Attempt 3    Wrong Password

Fail2Ban:
BAN 192.168.1.10
```

Result:

```
SSH Access Denied
```

Common Jails

SSH

```
[sshd]
enabled = true
```

Apache

```
[apache-auth]
enabled = true
```

Nginx

```
[nginx-http-auth]
enabled = true
```

FTP

```
[vsftpd]
enabled = true
```

Viva Questions

What is Fail2Ban?

A log-monitoring intrusion prevention tool that automatically bans IPs after repeated suspicious activities.

Which log does SSH jail monitor?

```
/var/log/auth.log
```

How to see banned IPs?

```
sudo fail2ban-client status sshd
```

How to unban an IP?

```
sudo fail2ban-client set sshd unbanip <IP>
```

Difference Between IDS and Fail2Ban

IDS	Fail2Ban
Detects attacks	Detects and blocks
Generates alerts	Creates firewall rules
Passive	Active prevention

Fail2Ban is an intrusion prevention tool that monitors log files and automatically bans IP addresses that perform repeated failed login attempts or other suspicious activities.